

Seat No : _____

No. of Printed Pages: 9

[31]

SARDAR PATEL UNIVERSITY
BCA Semester-5 External Examination, 2026
US05CBCA53 [Old Course] || SOFTWARE ENGINEERING



Date : 11/02/2026

Time : 10.00 a.m. – 01.00 p.m.

Total : 70 Marks

Q-1 Select the appropriate option for the following questions:

[10]

- 1) _____ is the collection of computer programs, procedures and data.
A. Hardware C. Network
B. Software D. Engineering
- 2) _____ requires major efforts.
A. Testing C. Coding
B. Maintenance D. Design
- 3) A high quality SRS reduces the development _____.
A. Time C. Cost
B. Customer requirements D. Quality
- 4) COCOMO stands for: _____.
A. Construction Cost Model C. Constructive Code Model
B. Constructive Cost Model D. Calculated Cost Model
- 5) Which of the following is a tool in design phase?
A. Abstraction C. Information Hiding
B. Refinement D. All of them
- 6) Structured design methodology tries to reduce _____.
A. Cost C. Cohesion
B. Time D. Coupling
- 7) A failure is produced only when there is a _____ in the system.
A. error C. fault
B. bug D. problem
- 8) Comments for a module are often called _____ for the module.
A. prologue C. information
B. message D. None of these
- 9) Which phase is required to understand the problem?
A. System Design C. Requirement Analysis
B. Coding D. Testing
- 10) _____ characteristic of SRS means the entire requirement denotes one interpretation.
A. Complete C. Unambiguous
B. Reliability D. Traceable

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE) [20]

- 1) Write a short note on maintenance phase.
- 2) Define: Software process and Software Engineering.
- 3) Explain advantages of spiral model.
- 4) List down at least 8 variables of COCOMO model.
- 5) Explain Partitioning.

[11]

(P.T.O.)

- 6) What do you mean by "Automated Cross Referencing"?
- 7) Differentiate between Functional and Object-oriented approaches.
- 8) Define: Module and Modular System.
- 9) Explain in brief Design walkthrough.
- 10) Write at least 2 differences between functional and structural testing.
- 11) What do you mean by information hiding?
- 12) Write the goal of coding.

- Q-3 A Explain Waterfall model. [06]
 B Explain characteristics of software process. [04]
- OR
- Q-3 A Explain Requirement Analysis and Testing phases. [06]
 B Explain iterative enhancement model. [04]
- Q-4 A Explain general characteristics of SRS. [05]
 B What is SRS? Explain needs of SRS. [05]
- OR
- Q-4 A Explain Single variable model for cost estimation. [05]
 B Explain the structure of SRS. [05]
- Q-5 A Discuss the design objectives in detail with proper illustrations. [06]
 B Write a short note on Coupling. [04]
- OR
- Q-5 A Explain levels of Cohesion. [06]
 B Explain verification techniques for system design. [04]
- Q-6 What is testing? Explain levels of testing in detail with diagrams. [10]
- OR
- Q-6 Explain coding approach and programming style. [10]